
BCS Maintenance

The BCS two-wheel tractor is an awesome tool that can be used in many ways, and reliability is the number one thing that market gardeners appreciate about it. You start using it, and it just works.

Here are a few pointers to keep in mind when operating a BCS. This tutorial is not a complete maintenance manual, but it should be enough to allow you to service and maintain your BCS properly.

The Engine (gas)

Oil

Oil is the lifeblood of your engine, and the more attention you pay to it, the better off your engine will be.

- We recommend that you check your oil EVERY time you fill your BCS with gas. It only takes a few seconds. To do so, level your tractor so that the base of the engine is parallel to the floor. You may have to place a block or something under your attachment to properly level it.
- If you use Honda's dipstick, do not screw it into the fill tube when you check the oil level. The "full" oil level is at the top thread of the oil fill tube, and the "add" oil level is at the bottom thread of the tube. Keep it full.
- After running the engine for the first time, make sure you change the oil after 5 to 8 hours of use. After that, change the oil every 40 hours of use. Any good quality 15W40 engine oil will work. In order to drain the old oil completely, run the engine long enough to warm the oil before draining it.

Air Filter

Air filter maintenance is also critical to your engine's performance.

-
- The foam prefilter that surrounds the paper cartridge should be removed and washed in warm, soapy water every 20 hours. Rinse and air dry. When dry, add a few tablespoons of engine oil to the foam element and squeeze to distribute it throughout the element.
 - At the same time, remove the paper cartridge and tap it a few times on a solid surface to remove loose dirt. Replace the paper cartridge every year.

Fuel

- Just like veggies – fresh is best. Do not buy and store more than a 60 day supply. Keep your jerrycan sealed and out of direct sunlight to maximize its storage life.
- Buy regular 87 octane gas but make sure that it contains no more than 10% ethanol.
- When transporting the tractor, remember to turn off the fuel valve -- the black slide lever located directly below the choke lever.

Spare Parts

- **Spark plug:** It doesn't need to be changed unless it's burned out, BUT it's always a good idea to have a spare one handy.
- **Recoil starter:** Rewinding the rope of your recoil starter is not difficult, but it can take some time (and practice). Since this always happens when tractor work is required in the field, it is often easier to simply replace the starter for immediate usage and fix it later.

The Clutch

For Standard Models, such as the 853

- The most common problem that we have seen over many years of use is when you squeeze the clutch and the tractor still moves forward. This is due to the cable stretching and requires that the clutch cable be adjusted and shortened to its original length. This most commonly

happens when reversing the handlebar and trying to drive it forward in reverse.

- The less common cable-related problem is when you release the clutch and lose power to the wheels and attachment. This indicates the cable is too tight and needs to be lengthened.
- Once adjusted, if the clutch still slips, this could mean that it's burned out. Burning out your clutch is the worst thing that can happen to your BCS. If this happens, we recommend sending it to a certified small engine repair mechanic, ideally the one who sold you the tractor.
- Whenever you are not going to use the tractor for a few days or longer, remember to squeeze the clutch lever and lock it in that position with the clip – just as you would if you were going to start the machine. This will prevent humidity from bonding the clutch cones together. If you fail to do so, you might not be able to squeeze the clutch lever after some point.

Best Practices

- Learn to “slip the clutch”. Sometimes, when shifting gears or engaging the PTO, you may find that the shift lever does not want to engage fully. This is because the gears in the transmission are not aligned properly. To engage fully, release the clutch lever very slowly while continuing to put pressure on the shifting rod. As soon as they begin to rotate, the pressure will pop the gears into place.
- When using power-intensive implements, like the flail mower, it's important to have the engine at 1/3 to 1/2 throttle while slowly letting out the clutch. After the clutch is fully released, you can (and should) accelerate the engine to maximum throttle.
- Never start and accelerate when your mower is filled with crop debris or residue. This will put an excessive load on the engine and cause premature wear of the clutch.
- When mowing and reaching the end of your row, it's best to NOT squeeze the clutch lever when turning. Keep the tool turning at full speed and make sure that it is at maximum revolutions per minute (RPM) before eating up more green manure.

For PowerSafe Models, such as the 749

- Replace the transmission oil filter and wipe off the magnetic prong after the first 30 hours of use. The filter and transmission oil should then be changed after every 100 hours of use.

The implements

- Inspect your implement before each use. Replace any worn or broken blades and make sure nothing is stuck in them (rope, metal, rocks, etc.).
- Most implements have a separate oil-filled transmission, so inspect for leaks and periodically check the oil level. Rule of thumb is that when you're replacing a completely worn out set of tines or blades, it's a good time to replace the oil. Either 80W-90 or 90W gear oil is fine.
- After the first 2-5 hours of operating a new implement, check all nuts and bolts for tightness. Continue to check periodically until there's no need.
- Grease any grease fittings twice a year and place some white grease on the quick coupler.
- To clean or power wash an implement, always disconnect it from the tractor. This will allow you to turn the implement on its side or back more easily and protects the tractor and engine.

Other Tractor Lubrication points

The rule of thumb for any equipment is: "If it moves, lubricate it!"

This applies to control levers, handlebar pivot and rotation points, cables, etc. So, if you want your investment in a walk-behind tractor to last as long as possible, keep it lubed up! We regularly use WD-40.

Troubleshooting Guide

Problem	Possible cause	Remedy
Grinding noise when trying to select a gear	The clutch cable is stretched so the clutch cones are not separating.	Shorten the cable by turning the adjusting tube outward. See video: https://www.youtube.com/watch?v=jMiFLtDmnb4
Unable to pull engine over	Check the dipstick. If overfilled, there may be petrol in the engine oil due to the fuel tap being left on during transportation.	Drain oil/gas mixture. Remove spark plug and air filter. Change the engine oil. Clean or change the air filter and spark plug.
No power to wheels or attached implement	The clutch cable is too tight so the clutch cones are not firmly contacting each other.	Lengthen the cable by turning the adjusting tube inward. See the video above.
Engine smoking / running rough	Contaminated fuel in the tank and carburetor	Drain and flush fuel from the tank. Replace with fresh gas and carburetor cleaning solvent. Contact the dealer if needed.
	Contaminated air filter	Clean or replace the air filter.